

Wiring — Meade LX200 GPS + ESP32

Same SH1106 OLED as AdhanOLED · RS232 jack (not HBX)

Product

Telescope: Meade LX200 GPS (Autostar II / Smart Mount base panel)
Serial jack: labeled RS232 (RJ11/RJ12) — never use HBX for this project
Screen: SH1106 128x64 I2C — same module/pins as Arduino/AdhanOLED
Baud: 9600 8N1 · Library: U8g2 (OLED_IS_SH1106=1 in config.h)

ESP32 GPIO map

ESP32 pin	Direction	Device	Device pin
GPIO21	SDA	OLED (AdhanOLED)	SDA
GPIO22	SCL	OLED (AdhanOLED)	SCL
3V3	power	OLED VCC	VCC
GND	ground	OLED GND	GND
GPIO17 (TX2)	out	MAX3232	T1IN
GPIO16 (RX2)	in	MAX3232	R1OUT
3V3 or 5V	power	MAX3232 VCC	(check module)
GND	ground	MAX3232 GND	GND
USB 5V	power	ESP32 board	—

Telescope base ports

Panel: OFF/ON · 12VDC · RETICLE · HBX · RS232 · RS232 · AUTOGUIDER ...
RS232 → YES — plug ESP32 MAX3232 cable here
HBX → NO — physical Autostar II coiled cable only
RETICLE / AUTOGUIDER → accessories only (not serial control)

Never connect RS-232 (± levels) directly to ESP32 GPIO — always use MAX3232.

Source: docs/WIRING.md · ESP32-LX200-Telescope-Controller

RJ11 pinout + signal path

Meade #507-style cable into LX200 GPS RS232 jack

RJ11 pinout (looking into jack, clip/tab down, pins 1→6 L→R)

RJ pin	Mount signal	Wire to MAX3232 RS232 side
1, 2, 6	unused	leave open
3	RX into telescope	T1OUT (from ESP TX / GPIO17)
4	GND	GND
5	TX from telescope	R1IN (to ESP RX / GPIO16)

Baud rate: 9600 · 8 data bits · no parity · 1 stop bit (8N1)

Signal path

ESP32 GPIO17  MAX3232 T1IN  (level shift)  T1OUT  RJ pin 3 (scope RX)

ESP32 GPIO16  MAX3232 R1OUT  (level shift)  R1IN  RJ pin 5 (scope TX)

ESP32 GND                                  RJ pin 4

If timeouts with a live mount: swap RJ pins 3↔5 once, confirm power LED on.

First power-on (start location)

Webpage forces Setup wizard (like Autostar II) until you enter:

- Latitude / longitude (where the telescope sits)
- Local date & time + UTC offset

Saved in ESP32 flash and pushed with :St :Sg :SL :SC :SG

Also see

[WIRING DIAGRAM.pdf](#) · [SYSTEM FLOWCHARTS.pdf](#) · [preview/AutostarII](#) [Preview.html](#)

GitHub: github.com/krisnoyarks-eng/ESP32-LX200-Telescope-Controller